

Geospatial applications – Web GIS

Term paper

by

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Introduction

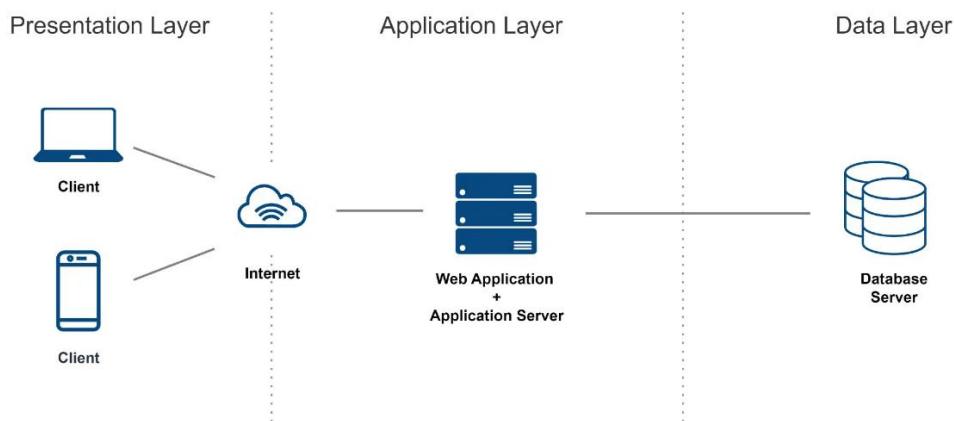
In the past, paper maps played a central role in how geographic information was accessed. However, with the rapid growth of information technology, geomatics engineers have moved from relying on paper maps to digital mapping and local GIS tools for storing and editing geographic data (Agrawal and Gupta, 2017). Simultaneously, web-based applications have seen significant growth, especially within commercial and land use management companies. These applications are interactive computer programs built using web technologies and designed to store and manipulate data to perform a specific task (Cameron and Ponton, 2002). These applications differ from traditional information systems due to their primary purpose and the scope of their audience reach (Taylor *et al.*, 2002).

The integration of geographic information systems with web-based technologies has led to the development of Web GIS. This technology combines the spatial analysis and data management capabilities of GIS with the accessibility and interactivity of the web. As a result, users can now visualise, analyse, and share geographic data in real time through a web browser, without needing specialised software. Web GIS has become an increasingly valuable tool across both public and private sectors, supporting faster decision-making and more efficient management of spatial resources online. This work aims to explore the concept of web GIS in more detail and introduce a web-based system called GeoForest, which was developed to demonstrate a practical application of this technology.

1. Traditional Web application architecture

A traditional website commonly works using the client-server (3-tier) Architecture, where the user interface is an application which communicates with the server through the internet (Lyu, Chu and Xue, 2017). Using the application programming interfaces (APIs), all the components can communicate together. Users interact with these tools within their web browsers. Everything is handled server-side rather than client-side. Server-side scripting languages (back-end), such as PHP or JavaScript for Node.js, perform computations on the server while client-side scripting languages (front-end), including JavaScript, execute on the user's browser. Combining back-end and front-end scripting is often recommended to balance performance, security and user experience (Lan, 2009). Traditional websites consist of:

- **A client**, which is the user accessing the application through a computer or smartphone
- **The web application**, which is a software built using a combination of multiple languages (HTML, CSS, JavaScript...) and accessible from a browser and listening to a port.
- **The server**, which represents a computer, is located somewhere which gives to the clients the information they need through several analyses.
- **The database**, which is the storage of the data used by the server.



All the elements are communicating through the World Wide Web, which is a network of hypertext documents that are linked together using Hypertext Transfer Protocol (HTTP).

1.1. Front End part

The client communicates with the server using a user interface, also called the front-end. The front-end is built using a combination of HTML, CSS and JavaScript.

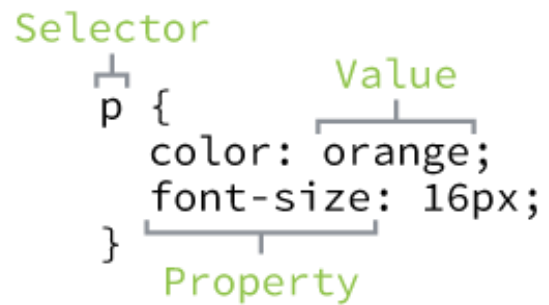
- **HTML** (hypertext Markup Language): a language which takes care of the structure and the content of the web page. It consists of the Head, which contains the HTML metadata, and the body, which holds the content of the HTML

```
<!DOCTYPE html>
<html>

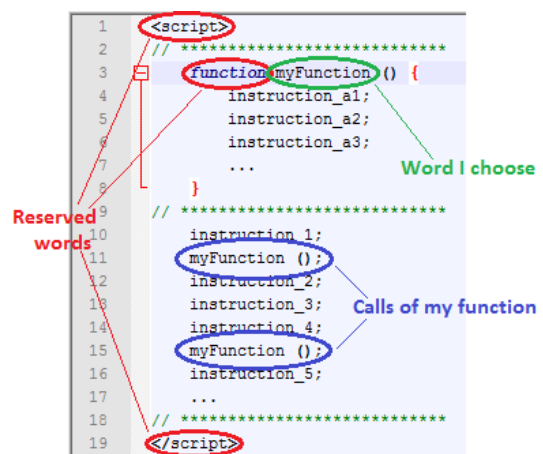
  <head>
    <title> Title here </title>
  </head>

  <body>
    Web page content goes here.
  </body>
</html>
```

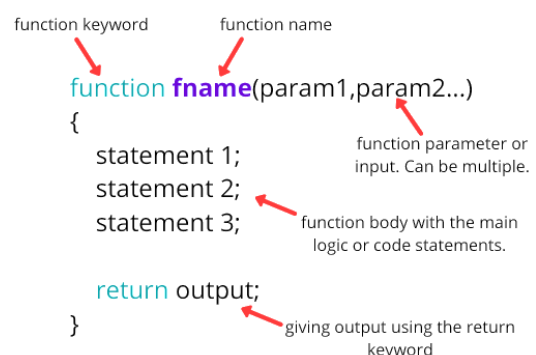
- **CSS** is an acronym for cascading style sheet. it takes care of the style of the user interface, also called the front end. The CSS refers to a style property assigned to each part of the web application visible to the users. It is made of the selector which define the specific CSS element, the properties which are the different style titles and the value of each style. Some producers like “W3School CSS” can furnish a wide range of style references that can be directly used from their library or online.



JavaScript code: it's a programming language which takes care of the interactivity and logic of the web application using the document object model (DOM), allowing the representation of the content of the web app. Each JavaScript file must start with `<Script>` and end with `</script>`. There are some reserved words to JavaScript that programmers cannot use in their code.



JavaScript works mostly with many functions consisting of statements used to perform specific tasks. Programmer can define their own functions using the function structure or used some JavaScript built-in functions (provided by JavaScript itself) like `alert ()` or `parseInt()`.



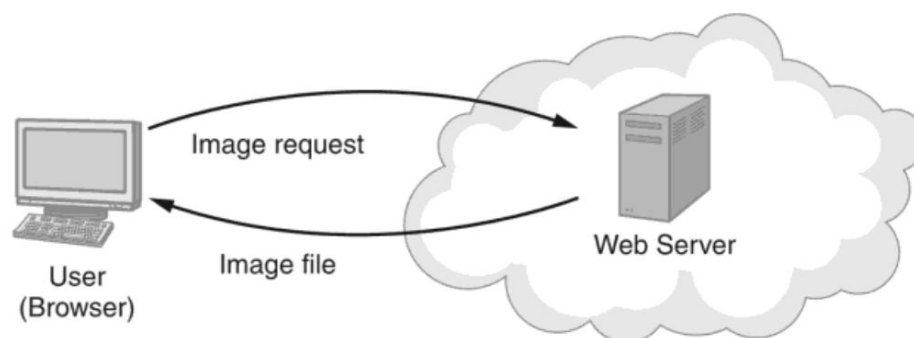
1.2. Back End part

PHP, for Hypertext Preprocessor, is one of the programming languages often used for back-end programming. It is used to implement server-side logic for dynamic web pages and

applications. With PHP, the Script/code will not be run in the browser but on the server. This allows calculation from the database, reading a different file format, accessing the database, accessing a file locally or starting another local application. There are just some specific rules to follow when using a specific server-side programming code, in PHP, for instance, variable names must start with the dollar sign (\$).

In addition to dynamic web applications, server-side programming is also used for security reasons by keeping sensitive operations and data on the server and therefore not accessible to the user/client. Some other back-end programming languages exist, like Node.js or C#, but PHP is still the most used with a very large community.

In terms of operability, developers create an HTML file, then use a special protocol called File Transfer Protocol (FTP) to upload files to the web server. To view those files, the user sends an HTML request through their browser to a web server, and the server checks if needed inside a specific database or processes it through a specific analysis in the back end and sends as a response to the user.



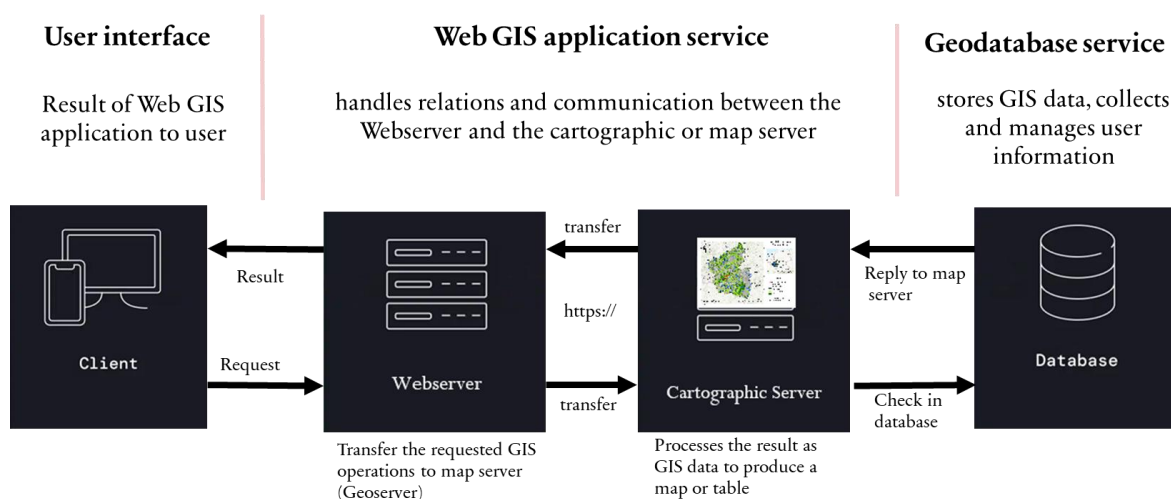
2. Web GIS application

2.1. Local GIS versus Web GIS

A local GIS is traditional software installed locally on a computer that aims to visualise, draw and analyse geospatial data, while a web GIS system uses the web/internet to make geospatial data and tools accessible to many users at the same time through the browser. So, it doesn't require individual installation and updates in the local computer like local GIS.

2.2. Functioning of a web GIS

A Web-GIS application is an application that uses World Wide Web technology to display, share geospatial information and manage it through various functionalities. In a web-GIS application, a client sends the request to the server and depending on the request, a web server or the map server, or both can respond to it. A traditional web application usually only contains a web server, but in the case of web-GIS, there is an additional server called a cartographic server or map server for spatial data. This server manages geospatial data, provides geospatial data-compatible services such as WMS and WFS, and is capable of performing GIS functions such as editing, routing, and tracking objects.



Giribabu et al. 2018

Like many web applications, a web-GIS works with a 3 Tier Architecture. In this architecture, most of the processing is done at the server side. When a client makes a request, the map server generates a response as a generated map, feature or coordinate using the database. To be able to proceed and manage geospatial data, in addition to the base JavaScript programming language, some mapping libraries can be used. A web GIS often use an interactive map, which needs a lot of JavaScript to control interaction, and the map is displayed in a map container. Events can also be added based on click or mouse over (Mover) on the map container or a specific feature. In a web GIS, it's also possible to draw new features using map canvas rendering and specific libraries. It's also important to notice that in web mapping, the good practice is to use EPSG4326 as the coordinate system.

2.3. JavaScript and Mapping Libraries

jQuery is a JavaScript library that offers many utilities, like events and effects and can work in combination with AJAX (a library for data loading). AJAX (can also be considered as a

JavaScript library) is mostly used for transferring data between the browser and server without the user having to reload the page every time. To do so, it uses the HTTP request and includes the response in the application without reloading the whole page. This can be done using a specific data format like JSON. jQuery can also be used to reduce the number of code lines needed to achieve a specific task. jQuery use “\$” to index an object.

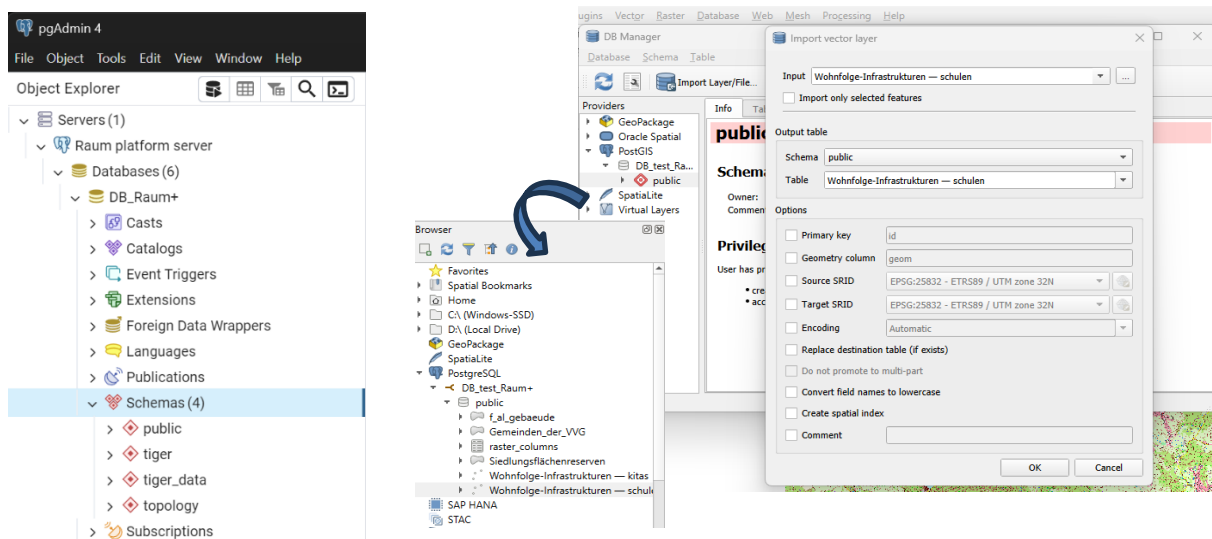
A mapping library is a JavaScript library that provides tools to display and interact with spatial data in the web browser. The particularity of mapping libraries is the ability to display, interact and analyse geospatial data in the form of WFS (Web Feature Service), WMS (Web Map Service) and JSON. The following table describes some of the mapping libraries mostly used.

Mapping library	Tool accessibility	Supported Data type	Advantages
Open layer	Open source	Many data source format is supported (WFS, WMS, GPX, KML, Geojson, xyz, vector tiles ...)	- Supported Operating systems: Windows, Linux and IOS - provide a full-featured mapping library - Colour compositing and raster operations - Snapping, geometries edition
Leaflet.js	Open source	Many data source format is supported (WFS, WMS, GPX, KML, Geojson, xyz, vector tiles ...)	- Supported Operating systems: Windows, Linux and IOS - Allow interactive Map - works efficiently across all major desktop and mobile platforms - Easy to use and can be extended with several plugins - Contain all mapping features
Turf.js	Open source	Use GeoJSON for all geographic data	- Supported Operating systems: Windows, Linux and IOS - Allow advanced geospatial analysis (tin, buffer, Bezier, Voronoi...)

2.4. The Database

A database is an organised collection of structured information, or data, typically stored electronically in a computer system. A database is usually controlled by a database management system (DBMS). PostgreSQL, for instance, as a DBMS is an open-source object-oriented database that contains the PostGIS extension, which enables the saving and the visualisation of geographic objects into the database. In a web GIS, a database is used to save data, edit data using SQL queries, implement user roles, and provide data for the map server. PostgreSQL can be managed with the command line or PgAdmin. The PostgreSQL database also contains schemes to manage access for the different user groups.

To import/export data in the geographic database, it can be connected to a local GIS software like QGIS using the PostGIS extension in QGIS, the database port, the database name and the password. Those data can then easily be properly used in the web application.



3. Case study of GeoForest web GIS.

GeoForest Cameroon is a web-based geographic information system (Web GIS) designed to visualise, explore and manage forest areas in Cameroon. The platform provides interactive mapping capabilities for different forest administrative types (managed forest, protected area and other environmental layers) using Leaflet.js, Turf.js, jQuery and W3 CSS reference. This platform is dedicated to being used by the forest administration of Cameroon (Ministry of Forestry) to easily know the location and different statuses of each Cameroon forest area.

- Technologies used

Components	Libraries/Tools	Purpose in the project
Base map and layers	Leaflet.js	Rendering interactive maps and vector layers
Geospatial analysis	Turf.js (online library)	Spatial masking
User interface design	W3.CSS	Styling and responsive layout
Scripting	JavaScript	Interactivity, data loading, and user interface logic
Data format	GeoJSON	Storing and visualising geospatial vector data
AJAX Data Loading	JQuery	Loading external GeoJSON files dynamically

The platform allows users to:

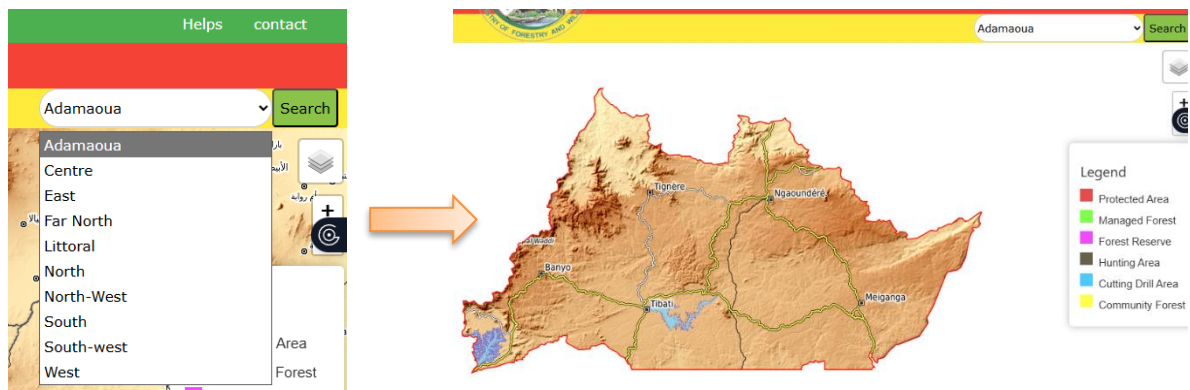
- Select and view regions of Cameroon.
- Based on the selected region, display specific forest layer (protected area, hunting area, forest reserve...)
- Using the Mask, the map display is focused on the selected region
- Interact with each spatial feature to view detailed attributes (name, management type, status...)
- Access helpful forms and contact forms directly within the interface.

- Main Features

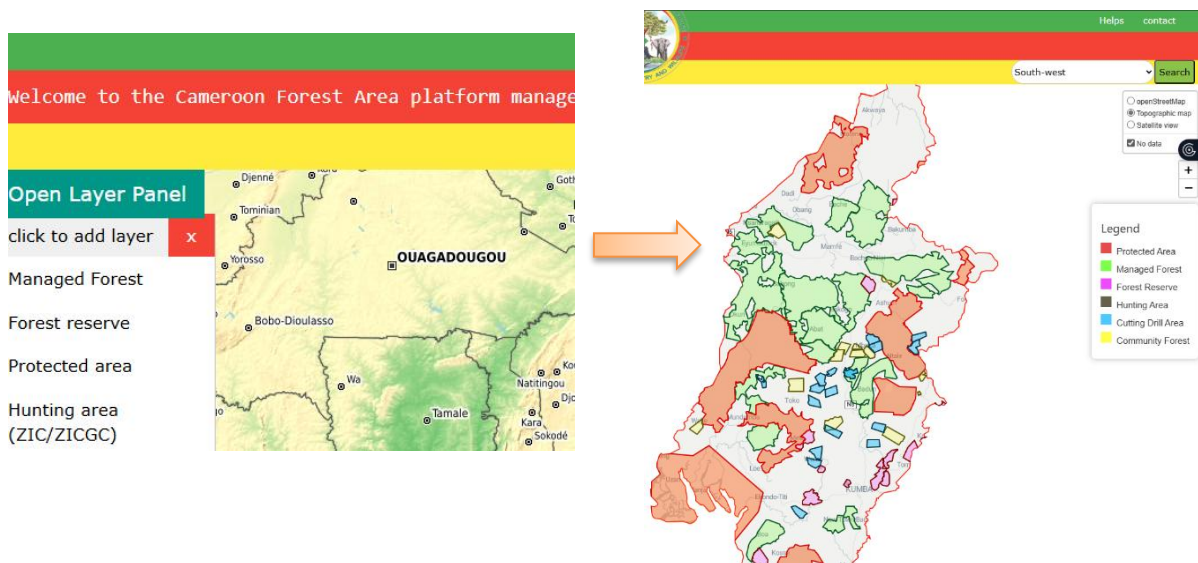
Map visualisation was built using Leaflet with OpenStreetMap (OSM) base layers and supports zooming, panning, and toggling between basemaps (topographic, satellite and OSM)



Region selection was implemented using a dropdown menu list, allowing users to select a region. Upon selection or when the search button is clicked, the chosen region is highlighted, and the rest of the map is masked out.



The Layer management tab (layer panel) contains clickable options (Managed Forest, Forest reserves, protected area...), and each option loads a specific GeoJSON file and applies a custom style and popups showing metadata (area, type, management status...)



The dynamic legend located at the top right corner of the map area displays the colour code and symbol for each layer. Also, the contact form and the Help form are used respectively to display contact information of the developer or the institution and allow users to describe issues they face.

Help form

First Name and Name

Email

You're face a problem with :

☒ Performance and technical issues
 ☐ Data and content Issues
 ☐ Operational issues
 ☐ Other

object

Report your issue(s)

send

Contact information

Name: Karel NIMPA

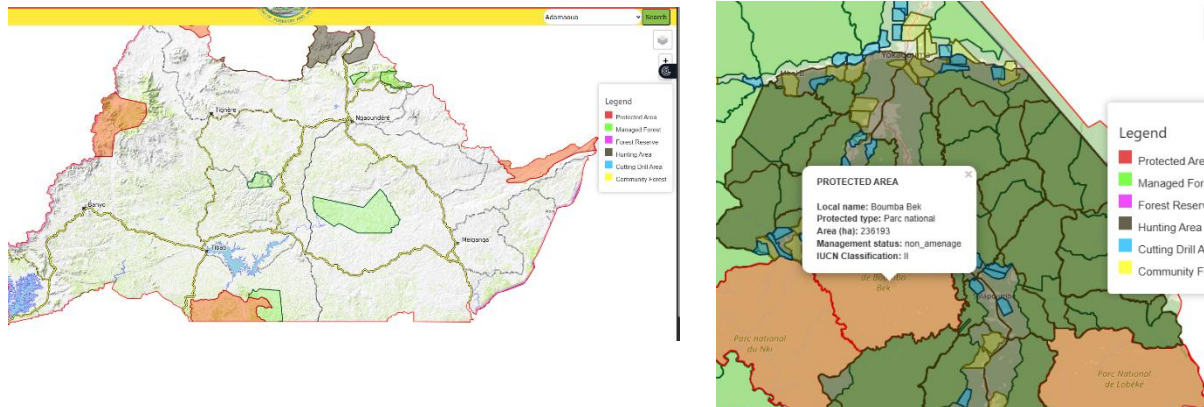
Email: s6kanimp@uni-trier.de

Institution: University of trier

Tel: +49 000 000 000

contact hours: Mon-Fr 10h-14h

In terms of user interaction, clicking on the map features opens pop-ups displaying detailed attributes. Each layer can be toggled on/off dynamically by clicking on the layer name in the layer panel. Finally, when selecting a region, the map auto-zooms on the selected region.



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